Home assignment - Senior Software Engineer - Nov 2026

Thanks for your time so far. As the next step, we'd like you to work on a short home assignment. It should take **up to 1 hour**. It's meant to give you a small taste of the kind of work we do, and to give us a concrete toy example to discuss together.

Tools

Use of AI and agentic development tools (Codex, Claude Code, Cursor etc.) is **allowed and recommended**. This reflects our development reality. If you don't have access to a coding agent, or need any help setting things up, just let us know and we'll sort it out.

The problem

You're given a time series of electricity consumption at 0.5-hour intervals, covering one month (attached: `consumption.csv`).

Electricity is purchased under the following pricing:

- Hours 17, 18, and 19 of each day (i.e. 17:00-20:00) cost 2 NIS/kWh
- All other hours cost 1 NIS/kWh.

You can buy a **contract** that changes the settlement. You buy a fixed amount (kWh per hour) that applies to every hour of the month, at 0.5 NIS/kWh. In any hour, consumption up to the contract amount is covered by the contract. Consumption above the contract amount is charged at the regular price for that hour. To clarify - you pay for a contract regardless of the actual usage [contract amount (kW) × total hours in month (h) × rate (NIS/kWh)].

Your task: find the optimal contract amount.

The audience

The result needs to be communicated to our business unit. They're not technical, and they'll need to understand and act on the outcome. Think about what they need from you, and in what form. Imagine that we provide the recommended outcome and they take it into the real world, make the actual deals and sign the contracts.

Deliverable

Share a repository (GitHub) with your work. Include whatever you think is relevant: code, results, and whatever you'd hand to the business unit.

Share with doron@ngrid.cloud and yuval@ngrid.cloud

What happens next

Once you are done, we will schedule a 1h frontal session that will conclude the professional assessment. We'll discuss:

- What you identified as the relevant primitives of the problem
- How you modeled it
- The approach you chose to solve it, and why
- How you communicated the result to the business unit, and why you chose that form

In all of the above - “you” refers to you and your agentic extensions :)

Assumptions and open questions you ran into are good material for the discussion, so feel free to note them.

Good luck, and reach out if anything is unclear.